# Econometrics Chapter 2: Conditional Expectations & Simple Linear Regressions

## (1) Data Modelling

Data Modelling

- attempt to model the population using a joint distribution function $f_{X,Y}(x,y)$
- discrete $X, Y$: $P(X=x, Y=y) = f_{X,Y}(x,y)$

continuous $X, Y$: $P(a \le X \le b, c \le Y \le d) = \int_a^b \int_c^d f_{X,Y}(x,y)\,dy\,dx$

## (2) Joint and Conditional - Discrete

| Y \ X | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 6 | 0 | 0 | 0 | 0 | $\frac{1}{20}$ |
| 5.5 | 0 | 0 | 0 | $\frac{1}{20}$ | $\frac{2}{20}$ |
| 5 | 0 | 0 | $\frac{1}{20}$ | $\frac{2}{20}$ | $\frac{1}{20}$ |
| 4.5 | 0 | $\frac{1}{20}$ | $\frac{2}{20}$ | $\frac{1}{20}$ | 0 |
| 4 | $\frac{1}{20}$ | $\frac{2}{20}$ | $\frac{1}{20}$ | 0 | 0 |
| 3.5 | $\frac{2}{20}$ | $\frac{1}{20}$ | 0 | 0 | 0 |
| 3 | $\frac{1}{20}$ | 0 | 0 | 0 | 0 |

Joint PDF

- Function to represent $P(X \cap Y)$

$f_{X,Y}(x,y) = P(X=x, Y=y)$

(i) Unconditional (Marginal) Distribution

- Finding the probabilities of each circumstances in X regardless of Y
  Finding the probabilities of each circumstances in Y regardless of X

| $x$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| mass value of X $f_X / P(X=x)$ | $\frac{4}{20}$ | $\frac{4}{20}$ | $\frac{4}{20}$ | $\frac{4}{20}$ | $\frac{4}{20}$ |

$E(X) = 3$

$Var(X) = 2$

| $y$ | 3 | 3.5 | 4 | 4.5 | 5 | 5.5 | 6 |
|---|---|---|---|---|---|---|---|
| mass value of Y $f_Y / P(Y=y)$ | $\frac{1}{20}$ | $\frac{3}{20}$ | $\frac{4}{20}$ | $\frac{4}{20}$ | $\frac{4}{20}$ | $\frac{3}{20}$ | $\frac{1}{20}$ |

$E(Y) = 4.5$

$Var(Y) = 0.625$

(ii) Conditional Distributions

- general formula: $P(Y=y|X=x) = \frac{P(Y=y, X=x)}{P(X=x)}$

or

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$$

(iii) Conditional Expectations

- $E(Y|X) = \sum y \, P_{Y|X}(y|x)$
- $Var(Y|X) = \sum (y - E(Y|Z))^2 . P_{Y|X}(y|x)$

Properties: [Treat Conditioning information as a constant]

- $E(aXY|X) = aXE(Y|X)$
- $Var(aXY|X) = a^2X^2 Var(Y|X)$
- $Var(aX|X) = 0$

## (iii) Covariance & Correlation

### Covariance

- unbounded
- not invariant to scale
- $\sigma_{X,Y} = Cov(X,Y) = E[(Y - E(Y))(X - E(X))]$

### Correlation

- bounded between -1 and 1
- invariant to scale
- $\rho_{X,Y} = Corr(X,Y) = \frac{E[(Y - E(Y))(X - E(X))]}{\sqrt{Var(X)}\sqrt{Var(Y)}}$

$$= \frac{Cov(X,Y)}{\sqrt{Var(X)}\sqrt{Var(Y)}}$$

### Sample $\{X_i, Y_i\}_{i=1}^n$

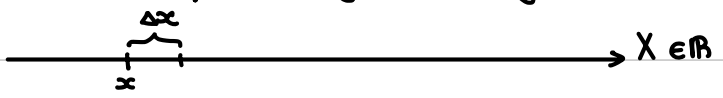
- $Cov = \hat{\sigma}_{X,Y} = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})$
- $Corr = \hat{\rho}_{X,Y} = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum(X_i - \bar{X})^2}\sqrt{\sum(Y_i - \bar{Y})^2}}$

### Properties

- $Var(aX + bY) = a^2 Var(X) + b^2 Var(Y) + 2ab\,Cov(X,Y)$
- $Cov(X,Y) = E(XY) - E(X)E(Y)$
- $Cov(X,X) = Var(X)$

## (3) Conditional Distribution - Continuous

- $f_X(x)$ — probability of being in the interval as $\Delta x \to 0$




$$E(X) = \int_{-\infty}^{\infty} x f_X(x) dx$$

$$E(Y) = \int_{-\infty}^{\infty} y f_Y(y) dy$$

## Marginal PDF

$$f_X(x) = \int_Y f_{X,Y}(x, y) dy$$

$$f_Y(y) = \int_X f_{X,Y}(x, y) dx$$

## Conditional Expectations

- $f_{X,Y}(x, y) = f_{Y|X}(y|x) f(x)$

  $= f_{X|Y}(x|y) f(y)$

- $E(Y|X=x) = \int_Y y f_{Y|X}(y|x) dy$
- $\text{Var}(Y|X=x) = \int_Y (y - E_{Y|X}(Y))^2 f_{Y|X}(y|x) dy$

## (4) Law of Iterated Expectations ☆

Law of Iterated Expectations

$$E_Y(Y) = E_X(E_{Y|X}(Y|X))$$

- the expected value of Y is equal to the expected value of its conditional expectation given X

i. Proving & Derivation [Discrete]

- allow $g(Y) = E(X|Y)$

$$E(E(X|Y)) = E(g(Y))$$
$$= \sum g(Y).P(Y)$$
$$= \sum(E(X|Y)).P(Y)$$
$$= E(X)$$ By total expectation theory

## ☆ 2 Important Results from the Law of Iterated Expectations ☆

No 1. If $E(Y|X) = c$

$E(Y) = c$ and $Cov(X,Y) = 0$

- Proof : $E(Y) = E(E(Y|X))$

$$= c$$

$$Cov(X,Y) = E(XY) - E(X)E(Y)$$
$$= E(X\,E(Y|X)) - cE(X)$$
$$= cE(X) - cE(X)$$
$$= 0$$

No 2. If $E(Y|X) = \beta_0 + \beta_1 X$

Implies that $\beta_0 = E(Y) - \beta_1 E(X)$ and $\beta_1 = \frac{Cov(X,Y)}{Var(X)}$

- Proof & Derivation:

$$E(Y) = E(E(Y|X))$$
$$= E(\beta_0) + \beta_1 E(X)$$
$$E(Y) = \beta_0 + \beta_1 E(X)$$
$$\beta_0 = E(Y) - \beta_1 E(X)$$

$$E(YX) = E(E(YX|X))$$
$$= E(XE(Y|X))$$
$$= E(X(\beta_0 + \beta_1(X))$$
$$= E(X\beta_0) + \beta_1 E(X^2)$$
$$= \beta_0 E(X) + \beta_1 E(X^2)$$

Given $\beta_0 = E(Y) - \beta_1 E(X)$

$$E(YX) = (E(Y) - \beta_1 E(X))E(X) + \beta_1 E(X^2)$$
$$E(YX) = E(Y)E(X) - \beta_1 (E(X))^2 + \beta_1 E(X^2)$$
$$\beta_1 = \frac{E(YX) - E(Y)E(X)}{E(X^2) - [E(X)]^2}$$
$$= \frac{Cov(X,Y)}{Var(X)}$$

# (5) Linear Regression

Linear Regression

- study of linear relationship between dependent variable Y and one or more explanatory variables X.
- two approach

Approach 1: Linear Approximation of a function $f(x)$

- $f(x) \approx f(a) + f'(a)(x-a)$
- Example: Allow $f(x) = \ln x$

$$\ln x_2 \approx \ln x_1 + \frac{1}{x_1}(x_2 - x_1)$$

$$\ln x_2 - \ln x_1 = \frac{x_2 - x_1}{x_1}$$

★Approach 2: Ordinary Least Squares (OLS)★

- Finding $\beta_0$ & $\beta_1$ to fit the model

$E(Y|X) = \beta_0 + \beta_1 X$

such that $\hat{\beta}_0$ and $\hat{\beta}_1$ is chosen to

Minimize $\sum(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2 = \sum(Y_i - \hat{Y})^2$

- Generalization:

$$\sum(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_{i1} - \hat{\beta}_2 X_{i2} - \ldots - \hat{\beta}_n X_{in})^2$$

# ★(6) Simple Linear Regression ★

(i) Population

- the purpose is to estimate $E(Y|X)$ for some population
- $E(Y|X) = \beta_0 + \beta_1 X$ (Simple Linear Regression Model)
  - define $e = Y - \beta_0 - \beta_1 X$
- Rewritten as:

  $Y = \beta_0 + \beta_1 X + \varepsilon$ whereby $E(\varepsilon|X) = 0$
  - $E(\varepsilon|X) = E(Y - \beta_0 - \beta_1|X)$

    $= E(Y|X) - \beta_0 - \beta_1 X$

    $= 0$
  - $E(\varepsilon|X) \Rightarrow E(\varepsilon) = 0 \Rightarrow E(\varepsilon X) = 0$
  - $E(Y|X) = \beta_0 + \beta_1 X$

    implies $\beta_1 = \frac{Cov(X,Y)}{Var(X)}$ and $\beta_0 = E(Y) - \beta_1 E(X)$

(ii) Sample $\{Y_i, X_i\}_{i=1}^n$

- assumptions:
  - random
  - iid
  - $\sum (X_i - \bar{X})^2 > 0$ there are variations in the $X_i$ observations
- model: $\widehat{E(Y|X)} = \hat{\beta}_0 + \hat{\beta}_1 X$

  $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$

  $Y_i = \hat{\beta}_0 + \hat{\beta}_1 X_i + \varepsilon_i$, $E(\varepsilon_i|X_i) = 0$ for $i = 1, \ldots, n$

Assumptions:

- $E(\varepsilon_i|X_1, \ldots, X_n) = 0$
- $E(\varepsilon_i \varepsilon_j|X_1, \ldots, X_n) = 0$ for $i, j = 1, \ldots, n$, $i \neq j$
- $Var(\varepsilon|X) = 0$
- $Cov(\varepsilon, X) = 0$

- Proof of $Cov(\varepsilon, X) = 0$

  $E(\varepsilon X|X) = 0$ therefore $E(\varepsilon X) = 0$

  $Cov(\varepsilon, X) = E(\varepsilon X) - E(\varepsilon)E(X) = 0$

## (iii) Estimation by OLS ★

- Given any potential estimates $\hat{\beta}_0$ and $\hat{\beta}_1$
- Define:
  - Fitted Values: $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$
  - Residuals: $\hat{\varepsilon}_i = Y_i - \hat{Y}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i$
  - SSR $= \sum (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2$
  - OLS $= \underset{\hat{\beta}_0^{ols}\ \hat{\beta}_1^{ols}}{\operatorname{argmin}} \sum (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2$

- First Order Condition

$$\underset{\hat{\beta}_0^{ols}\ \hat{\beta}_1^{ols}}{\operatorname{argmin}} \sum (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2$$

$$\frac{\partial SSR}{\partial \hat{\beta}_0} = -2\sum (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0$$

$$\frac{\partial SSR}{\partial \hat{\beta}_1} = -2\sum X_i (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0$$

Solving:

1. $\hat{\beta}_0^{ols} = \bar{Y} - \hat{\beta}_1^{ols} \bar{X}$

Proof: $-2\sum (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0$

$$\sum Y_i - \sum \hat{\beta}_0 - \hat{\beta}_1 \sum X_i = 0$$

$$\sum Y_i - n\hat{\beta}_0 - \hat{\beta}_1 \sum X_i = 0$$

$$\frac{1}{n}\bar{Y} - n\hat{\beta}_0 - \frac{1}{n}\hat{\beta}_1 \bar{X} = 0$$

$$n\hat{\beta}_0 = \frac{1}{n}\bar{Y} - \frac{1}{n}\hat{\beta}_1 \bar{X}$$

2. $\hat{\beta}_1^{ols} = \dfrac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sum(X_i - \bar{x})X_i}$

Proof: $-2\sum X_i(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0$

$\sum X_i(Y_i - (\bar{Y} - \hat{\beta}_1 \bar{X}) - \hat{\beta}_1 X_i) = 0$

$\sum X_i[(Y_i - \bar{Y}) + \hat{\beta}_1(\bar{X} - X_i)] = 0$

$\sum X_i(Y_i - \bar{Y}) + \hat{\beta}_1 \sum X_i(\bar{X} - X_i) = 0$

$-\hat{\beta}_1 \sum X_i(\bar{X} - X_i) = \sum X_i(Y_i - \bar{Y})$

$\hat{\beta}_1 \sum X_i(X_i - \bar{x}) = \sum X_i(Y_i - \bar{Y})$

$\hat{\beta}_1^{ols} = \dfrac{\sum X_i(Y_i - \bar{Y})}{\sum X_i(X_i - \bar{x})}$

$= \dfrac{\frac{1}{n-1}\sum(x_i - \bar{x})(Y_i - \bar{Y})}{\frac{1}{n-1}\sum(x_i - \bar{x})^2}$

$= \dfrac{\text{Sample Cov}(X_i, Y_i)}{\text{Sample Var}(X_i)}$

- Second Order Condition (using directional vectors)

$\dfrac{\partial^2 SSR}{\partial \hat{\beta}_0^2} = 2n$

$\dfrac{\partial^2 SSR}{\partial \hat{\beta}_1^2} = 2\sum X_i^2$

$\dfrac{\partial^2 SSR}{\partial \hat{\beta}_0 \hat{\beta}_1} = 2\sum X_i$

(iv) Evaluating the estimators $\hat{\beta}_1^{ols}$

a. Unbiasness of $\hat{\beta}_1^{ols}$

$$\hat{\beta}_1^{ols} = \frac{\sum(X_i - \bar{X})Y_i}{\sum(X_i - \bar{X})X_i}$$

$$= \frac{\sum(X_i - \bar{X})(\beta_0 + \beta_1 X_i + \varepsilon_i)}{\sum(X_i - \bar{X})X_i}$$

$$= \frac{\beta_0 \sum(X_i - \bar{X}) + \beta_1 \sum(X_i - \bar{X})X_i + \sum(X_i - \bar{X})\varepsilon_i}{\sum(X_i - \bar{X})X_i}$$ ($\beta_0 \sum(X_i - \bar{X}) = 0$)

$$= \beta_1 + \frac{\sum(X_i - \bar{X})\varepsilon_i}{\sum(X_i - \bar{X})X_i}$$

$$E(\hat{\beta}_1^{ols} | X_1, \ldots, X_n) = E(\beta_1) + \frac{\sum(X_i - \bar{X}) E(\varepsilon_i | X_1, \ldots, X_n)}{\sum(X_i - \bar{X})X_i}$$ ($E(\varepsilon_i | X_1, \ldots, X_n) = 0$)

$$= \beta_1$$

$\therefore \hat{\beta}_1^{ols}$ is unbias for $\beta_1$

$\therefore$ Therefore, we show that $E(\hat{\beta}_1^{ols}) = \beta_1$

b. Consistency of $\hat{\beta}_1^{ols}$

Two Arguments:

1. $$\hat{\beta}_1^{ols} = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sum(X_i - \bar{X})X_i} = \frac{\frac{1}{n}\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\frac{1}{n}\sum(X_i - \bar{X})X_i} \xrightarrow{p} \frac{Cov(X,Y)}{Var(X)} = \beta_1$$

2. $$\hat{\beta}_1^{ols} = \beta_1 + \frac{\sum(X_i - \bar{X})\varepsilon_i}{\sum(X_i - \bar{X})X_i}$$

$$= \beta_1 + \frac{\frac{1}{n}\sum(X_i - \bar{X})\varepsilon_i}{\frac{1}{n}\sum(X_i - \bar{X})X_i} \xrightarrow{p} \beta_1 + \frac{Cov(\varepsilon, X)}{Var(X)}$$ ($Cov(\varepsilon, X) = 0$)

$$= \beta_1$$

$\therefore$ Conclusion: Empirical Model: $\hat{Y} = \hat{\beta}_0^{ols} + \hat{\beta}_1^{ols} X_i$ is unbiased & consistent for $E(Y|X)$

## (v) OLS Standard Errors

a. $Var(\varepsilon_i | X_1, \ldots, X_n) = \sigma^2$ [Homoskedasticity]

- noise of data does not depend on any of the X observations
- Note: Heteroskedasticity — variance of error increases in X

b. $$Var(\hat{\beta}_1^{ols} | X_1, \ldots, X_n) = \frac{\sum (X_i - \bar{x})^2 Var(\varepsilon_i | X_1, \ldots, X_n)}{(\sum (X_i - \bar{x})^2)^2}$$

$$= \frac{\sum (X_i - \bar{X})^2 \sigma^2}{(\sum (X_i - \bar{X})^2)^2}$$

$$= \frac{\sigma^2}{\sum (X_i - \bar{x})^2}$$

- Implication: $\hat{\beta}_1$ has a smaller variance if,
  - $\sigma^2$ is smaller (data is less noisy)
  - n is larger
  - more variation in $X_i$ sample
- Estimating the variance
  - (i) Estimate for $\sigma^2$ is needed

    $\hat{\sigma}^2 = \frac{1}{n-2} \sum \hat{\varepsilon}_{i,ols}^2$ [only for homoskedastic case]
  - (ii) Variance of $\hat{\beta}_1$

    $\hat{\beta}_1 = \frac{\hat{\sigma}^2}{\sum (X_i - \bar{x})^2}$
  - (iii) Standard error of $\hat{\beta}_1$

    $$s.e(\hat{\beta}_1) = \sqrt{\frac{\hat{\sigma}^2}{\sum (X_i - \bar{X})^2}}$$

## (vi) Heteroskedastic Robust Standard Errors

$$\widehat{Var}_{HC}(\hat{\beta}_1^{ols}) = \frac{\sum (X_i - \bar{X})^2 \hat{\varepsilon}_{i,ols}^2}{(\sum (X_i - \bar{x})^2)^2} \xrightarrow{p} Var(\hat{\beta}_1^{ols})$$

- White/HC variance estimator allow for an estimation that converges in probability

$$s.e_{HC}(\hat{\beta}_1) = \sqrt{\frac{\sum (X_i - \bar{x})^2 \hat{\varepsilon}_{i,ols}^2}{(\sum (X_i - \bar{x})^2)^2}}$$

# (7) Problems Establishing Casual Interpretations

## 1. Omitted Variables

- happens when Y is dependent on two or more explanatory variable, and these explanatory variables are correlated
- True model: $Y_i = \beta_0 + \beta_1 X_i + \beta_2 Z_i + u_i$

  Omitted model: $Y_i = \alpha_0 + \alpha_1 X_i + \varepsilon_i$ , $\varepsilon_i = u_i + \beta_2 Z_i$

  Assume $Z_i = X_i + v_i$ , $v_i \sim$ Normal $(0, \sigma_v^2)$
- Solving Omitted Model:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

$$\varepsilon = u_i + \beta_2 Z_i$$

$\therefore E(\varepsilon | X) \neq 0$ , $\hat{\beta}_1$ in simple linear regression of Y on X is bias for $\beta_1$

## 2. Simultaneity Bias

- occurs when explanatory variable and dependent variable Y are determined at the same time, so that X is correlated with the error term in the regression
- cause and effect run both ways, making it impossible for OLS to isolate true direction of causality
- Example: Market for a good

$Q_t^d = \delta_0 + \delta_1 P_t + \varepsilon_t^d$ (Demand Eq$^n$, $\delta_1 < 0$)

$Q_t^s = \alpha_0 + \alpha_1 P_t + \varepsilon_t^s$ (Supply Eq$^n$, $\alpha_1 > 0$)

$Q_t^s = Q_t^d$ (Market Clearing)

- Solving:

$$\delta_0 + \delta_1 P_t + \varepsilon_t^d = \alpha_0 + \alpha_1 P_t + \varepsilon_t^s$$

$$P_t = \frac{\alpha_0 - \delta_0}{\delta_1 - \alpha_1} + \underbrace{\frac{\varepsilon_t^s - \varepsilon_t^d}{\delta_1 - \alpha_1}}_{\text{shocks/noise terms}}$$

$$Q_t = \left(\delta_0 + \frac{\alpha_0 - \delta_0}{\delta_1 - \alpha_1}\right) + \frac{\delta_1 \varepsilon_t^s - \alpha_1 \varepsilon_t^d}{\delta_1 - \alpha_1}$$

Extra: $\text{Var}(P_t) = \frac{\sigma_s^2 + \sigma_d^2}{(\delta_1 - \alpha_1)^2}$

$\text{Cov}(P_t, Q_t) = \frac{\delta_1 \sigma_s^2 + \alpha_1 \sigma_d^2}{(\delta_1 - \alpha_1)^2}$

$\hat{\beta}_1 \xrightarrow{p} \frac{\text{Cov}(Q_t, P_t)}{\text{Var}(P_t)} = \frac{\delta_1 \sigma_s^2 + \alpha_1 \sigma_d^2}{\sigma_s^2 + \sigma_d^2}$

- Regressor $P_t$ is endogenous, $E(\varepsilon | P_t) \neq 0$ prices are correlated to $\varepsilon$

  Regressor $P_t$ will be biased and inconsistent

$\therefore Q_t$ and $P_t$ is both endogenous

$\therefore \hat{\beta}_1$ is neither PES or PED

(8) Sampling Problems – problems that make your sample not representative of the population

1. Measurement Error

- when X can only be observed with error

$X^* = X + u$, measurement error $u$ is independent of $X$

- Model: $Y = \beta_0 + \beta_1 X^* + \varepsilon$

$= \beta_0 + \beta_1 X - \beta_1 u + \varepsilon$

$= \beta_0 + \beta_1 X + v$, $v = -\beta_1 u + \varepsilon$ [Composite error term]

- $E(v|X) \neq 0$ since $v$ is dependent on $\beta_1$

$\therefore \beta_1$ is therefore bias

If $\beta_1 > 0$, pushes $v$ downwards

If $\beta_1 < 0$, pushes $v$ upwards

2. Truncated Sampling – Non Random Sampling

- induces correlation between $\varepsilon$ and $X$

Soln: Limited Dependent Variance

- large $X_i$ with large positive error — +ve correlation btwn $X_i$ and $\varepsilon_i$
- large $X_i$ with negative error — -ve correlation btwn $X_i$ and $\varepsilon_i$

3. Specification issue

- True Model: $E(Y|X) = \beta_0 + \beta_1 X + \beta_2 X^2$

Model Utilized: $E(Y|X) = \beta_0 + \beta_1 X$

- biased but smaller SE